



Stable performance for pouch-type all-solid-state batteries enabled by current collector with optimized primer layer

Hyeonseong Oh^{a,b}, Jun Tae Kim^a, Hyeon-Ji Shin^a, A-Yeon Kim^a, Cheol Bak^c, Sang-Ok Kim^{a,d}, Kyung Yoon Chung^{a,d}, Junyoung Mun^{a,e}, Jongsoon Kim^{e,f}, Yong Min Lee^{b,*}, Sang-Young Lee^{b,*}, Hun-Gi Jung^{a,d,e,f,**}

^a Energy Storage Research Center, Sustainable Energy Research Division, Korea Institute of Science and Technology, Seoul 02792, Republic of Korea

^b Department of Chemical and Biomolecular Engineering, Yonsei University, Seoul 03722, Republic of Korea

^c Department of Energy Science and Engineering, DGIST, Daegu 42988, Republic of Korea

^d Division of Energy and Environment Technology, KIST School, Korea University of Science and Technology, Seoul 02792, Republic of Korea

^e KIST-SKKU Carbon-Neutral Research Center, Sungkyunkwan University, Suwon 16419, Republic of Korea

^f Department of Energy Science, Sungkyunkwan University, Suwon 16419, Republic of Korea

ARTICLE INFO

Keywords:

All-solid-state battery
Sulfide solid electrolyte
Pouch-type cell
Primer layer
Cu corrosion
External pressure

ABSTRACT

Sulfide-based all-solid-state batteries (ASSBs) are advancing beyond solid electrolyte development to focus on composite electrode design and scalability for commercialization. Scaling from laboratory prototypes to pilot-scale production of large cells with high-energy density and high-performance ASSBs introduces new challenges. Large-scale electrode development necessitates selection of suitable polymeric binders that are compatible with sulfide electrolytes and exhibit strong binding forces for enhanced longevity. In this regard, we used rubber- and styrene-based polymeric binders in ASSBs and introduced a primer layer on the current collector, combining carbon conductive agents and polyvinylidene fluoride, to improve adhesion to the current collector. This primer layer reduces the binder content, which is an inactive component in the electrode, thereby diversifying Li-ion and electron conduction pathways and enhancing the ionic and electronic conductivity of the composite electrode. Furthermore, the primer layer not only prevents direct contact between the sulfide-based solid electrolyte and the Cu current collector but also blocks interactions between the Cu current collector and vaporized sulfur. By acting as a passivation layer, it effectively suppresses Cu corrosion. Consequently, a pouch-type full cell incorporating a primer layer demonstrated improved initial capacity and higher Coulombic efficiency under non-pressurized conditions. Specifically, the cell retained 80.7 % of its capacity after 100 cycles, demonstrating improved performance compared to uncoated full cells (64.8 %).

1. Introduction

As the transition from internal combustion engines to electric propulsion systems has become a reality, the growing demand for high-performance electric vehicles has driven the development of next-generation batteries beyond liquid electrolyte-based lithium-ion batteries (LIBs)[1]. However, safety concerns associated with the use of organic liquid electrolytes in LIBs remain a significant challenge[2–4]. In response, all-solid-state batteries (ASSBs) have emerged as promising next-generation batteries, with substantial progress made over the past

decade in enhancing ionic conductivity and cell performance through the development of various solid electrolytes[5–8]. In particular, ASSBs with sulfide-based solid electrolytes (SSEs) have gained significant attention, demonstrating ionic conductivity comparable to that of liquid electrolytes while offering the advantages of cold pressing[9–14]. To commercialize sulfide-based ASSBs, ongoing research aims to ensure consistent ionic conductivity under harsh conditions, expand Li-ion pathways at interfaces, stabilize interfaces at elevated voltages, and develop novel anode systems including lithium metal[15–22]. One notable advantage of transitioning from LIBs to ASSBs is the ability to

* Corresponding author.

** Corresponding author at: Energy Storage Research Center, Sustainable Energy Research Division, Korea Institute of Science and Technology, Seoul 02792, Republic of Korea.

E-mail addresses: syleek@yonsei.ac.kr (S.-Y. Lee), hungi@kist.re.kr (H.-G. Jung).

<https://doi.org/10.1016/j.mser.2025.100970>

Received 15 December 2024; Received in revised form 5 February 2025; Accepted 24 February 2025

0927-796X/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

adapt classical LIB active materials (AMs) with minimal modifications. Additionally, the slurry-casting process for manufacturing large-area ASSB electrodes can proceed similarly by simply incorporating SSEs into the powder mixing process. However, SSEs are chemically unstable, which limits the use of conventional polar solvents, such as N-methyl-2-pyrrolidinone (NMP) and ethanol[23,24]. This limitation fundamentally affects the choice of polymeric binder, rendering traditional binders like polyvinylidene fluoride (PVDF) unsuitable[25]. Alternative binders such as nitrile butadiene rubber (NBR) and styrene-butadiene-styrene (SBS), soluble in several nonpolar or less polar solvents (e.g., anisole, xylene, and acetonitrile), exhibit weaker binding strength compared to classical binders (e.g., PVDF) due to their lack of polar groups. Consequently, the reduced interfacial adhesion between the current collector and electrode necessitates the addition of large amounts of binder to compensate for this low binding strength; however, this obstructs the Li-ion and electron movement, thereby degrading cell performance[26–28]. This study introduces a primer layer, comprising PVDF with a strong binding force and carbon conductive agents, between the electrode and current collector. This approach simultaneously enhances the interfacial adhesion and increases the surface roughness of the current collector for improved electrode bonding area, preventing electrode delamination and facilitating a more stable electrode–current collector interface. The introduction of the primer layer allowed for a significant reduction in binder content within the composite electrode (up to 80 wt%), diversifying the Li-ion and electron conduction pathways. This led to improved electrochemical performance, including enhanced initial discharge capacity and cyclability. Furthermore, the primer layer functions as a passivation layer, effectively mitigating reactions between Cu and S at the anode. The proposed primer system can be applied across both positive and negative electrodes, resulting in significant performance improvements for full cells. Its ability to provide sufficient adhesion under low external pressure is particularly advantageous for large-area ASSBs, demonstrating excellent electrochemical performance.

2. Materials and methods

2.1. Material preparation

Li₆PS₅Cl (LPSCl) was synthesized via solid-state mechanochemical reactions. Stoichiometric quantities of precursors, Li₂S (99.98 %, Sigma-Aldrich), P₂S₅ (99 %, Sigma-Aldrich), and LiCl (99.98 %, Sigma-Aldrich), were combined in a zirconia jar with zirconia balls and subsequently milled at 510 rpm for 60 h using a planetary ball mill (Pulverisette 6, Fritsch). The resulting glass powders underwent annealing in a sealed quartz tube within an Ar-conditioned furnace at 550°C for 5 h. The wet-milling method was used to prepare small-sized electrolyte particles to be used in the composite electrode. Zirconia balls, LPSCl powder, and mixed solvent of heptane and dibutyl ether were added to the zirconia jar and processed using a ball mill (Pulverisette 7, Fritsch). The volume ratio of the mixed solvent of heptane and dibutyl ether is 8:1. The ionic conductivity of the Small-sized Li₆PS₅Cl (S-LPSCl) is 1.69 mS/cm, and the electronic conductivity is 4×10^{-5} mS/cm. The 50 % cumulative population (D₅₀) is 0.91 μm (Figure S1). The milled LPSCl powder was subsequently vacuum-dried overnight in a Büchi oven. LiNbO₃ coating on LiNi_{0.82}Co_{0.13}Mn_{0.05}O₂ (NCM82, SMLAB) was accomplished by dissolving Li metal (Honjo Metal Co., Ltd.) and Niobium ethoxide (99.95 %, Sigma-Aldrich) in ethanol at stoichiometric proportions. NCM powder was added to the solution, followed by vigorous stirring for 30 min and solvent evaporation. The powder was then sintered at 400°C under an O₂ flow. Figure S1 shows the particle size analysis results for the particle distribution of NCM82 and graphite.

2.2. Primer and electrode fabrication

Primer layer deposition on the current collector utilized the doctor

blade technique, with different types of carbon conductive agents (graphene, super P) and distinct carbon to PVDF (solef 5130, Solvay) ratios for the anode and cathode. Given the difference in electronic conductivities of Cu and Al current collectors, solution compositions were tailored accordingly. After adding carbon conductive agents, the solutions were stirred for 1 h, and the carbon was uniformly dispersed using a horn sonicator. Respective primer solutions were coated onto bare Al (Al C.C) and Cu current collectors (Cu C.C), subsequently designated as primer@Al and primer@Cu, respectively. The electrodes were fabricated via the doctor blade method using a premixed powder and binder solution. The mixing of active materials (NCM, graphite), S-LPSCl, and Super P was performed using a mixer mill (MM400, Retsch). After adding SUS balls and powders in a grinding jar, they were mixed at 20 Hz for 20 minutes while being careful not to physically damage the particles. The pristine cathode composition was 80:17:1.5:1.5 (NCM82:LPSCl:super P:NBR) by weight, and the composition of the cathode with a low binder content (LBC) was 80:17:1.5:0.2. The pristine anode composition was 56:40:1:3 (graphite:LPSCl:super P:SBS), and the composition of the anode with LBC was 56:40:1:2. Both SBS and NBR binders were dissolved in anisole, with loading levels adjusted through additional anisole incorporation into the slurry. The mixed powder and binder solution were mixed at 2000 rpm for 6 minutes using a thinky mixer (ARE-310, THINKY corp.). The prepared slurries were then coated onto their respective current collectors to fabricate the electrodes. Both the cathode and anode electrodes were first dried for 30 minutes on a 60°C hot plate in the glove box immediately after casting. Additionally, vacuum drying at room temperature was performed for 10 hours in a chamber connected to the glove box.

2.3. Electrochemical characterization

For the half-cell tests, the vacuum-dried electrodes were punched into 12-mm diameter discs and used as the working electrodes. First, 110 mg of the LPSCl SSE powder pressed at 170 MPa into a 13-mm diameter mold. Subsequently, the punched electrode was placed on one side of the SSE powder pellet and pressed at 370 MPa. An Li–In alloy was added on the opposite side of the working electrode and pressed at 200 MPa. For the full-cell tests, an SSE film was prepared by mixing LPSCl powder with NBR in a 98:2 ratio. The SSE film used in the pouch-type full cell was prepared in the form of a free-standing film. LPSCl and NBR were mixed at a weight ratio of 98:2 at 2000 rpm for 6 minutes using a Thinky mixer in a glove box. The mixed solution was then cast on a polyethylene terephthalate (PET) film and dried at room temperature without using a hot plate. After the solvent was completely evaporated, vacuum drying was performed at room temperature for 10 hours in a chamber connected to the glove box. Afterwards, the SSE film was peeled off from the PET film in the glove box to obtain a free-standing film. In the actual pouch cell, two sheets of the cut SSE film were overlapped and used as a membrane. This is to prevent short circuit, and the thickness of the combined two sheets was 80 μm after warm isostatic pressure (WIP), as shown in Figure S2. The ionic conductivity of the large-sized LPSCl (L-LPSCl) used in the fabrication of the electrolyte film is 4.17 mS/cm, and the ionic conductivity of the electrolyte film used in the actual pouch-type full cell is 2.38 mS/cm. The results of each EIS measurement are shown in Figure S3. The cathode film was cut to a 40 × 30 mm² size, whereas the SSE and anode films were cut to a 42 × 32 mm² size. Al and Ni tabs were ultrasonically welded onto the cathode and anode, respectively. The cathode, SSE film, and anode were sequentially stacked into an Al pouch to assemble the full cell. Finally, the packaged pouch-type full cell was compressed at 480 MPa and 60°C using a WIP instrument (Ilshin autoclave). The active material mass loading of the cathode is 16.5–17.1 mg/cm² and the thickness of cathode is approximately 80 μm, excluding the Al current collector. The active material mass loading of the anode is 9.5–11 mg/cm² and the thickness is approximately 85–110 μm, excluding the Cu current collector (Figure S4). The N/P ratio was maintained at 1.15–1.2, with 1 C

= 180 mA g⁻¹. Galvanostatic cycling tests were conducted on coin-type half cells for cathode comparison and pouch-type full cells using a battery cycler (BCS-805, BioLogic) in the constant current (CC)/constant voltage mode for charging and the CC mode for discharging at 30°C in the voltage ranges of 2.1–3.7 V (vs. Li⁺/Li-In) and 2.7–4.3 V (vs. Li⁺/Li), respectively. Additionally, for anode comparison, galvanostatic cycling tests were conducted on coin-type half cells using a battery cycler (WBCS3000, WonATech) in CC mode during charge and discharge in the voltage range of -0.59–0.9 V (vs. Li⁺/Li-In) at 30°C. All cells were operated without applying external pressure. For cyclic voltammetry (CV) measurements, symmetric cells were manufactured using the current collectors to be evaluated. A mixed slurry of LPSCl (95 wt%) and NBR (5 wt%) was cast onto the current collector, punched, and used as an electrode for the symmetric cell. Similar to the half-cell assembly, 110 mg LPSCl powder was pressed into a 13-mm diameter mold at 170 MPa. Subsequently, the punched electrode was placed on both sides of the SSE powder pellet and pressed at 370 MPa. Electrochemical tests were conducted using a battery cycler (BCS-805, BioLogic) in potential windows of 2.7–4.3 V (vs. Li⁺/Li) for Al C.C and primer@Al and 0.01–1.5 V (vs. Li⁺/Li) for Cu C.C and primer@Cu. For all CV measurements, the scan rate was 0.1 mV/s, and tests were conducted for 50 cycles. The galvanostatic intermittent titration technique (GITT) measurements were performed using Li metal as the counter electrode in all cells, with a battery cycler (BCS-805, BioLogic). During the first cycle, a 0.3-C pulse was applied for 10 min, followed by 2-h rest. The contact area between the AM of the electrode and SSE was calculated using Eq. 1 based on the GITT measurements:

$$D = \frac{4}{\pi\tau} \left(\frac{m_{AM} \cdot V_M}{M_{AM} \cdot S} \right)^2 \cdot \left(\frac{\Delta E_s}{\Delta E_t} \right)^2, \quad (1)$$

where D represents the chemical diffusion coefficient of the AM (NCM82 or graphite), S represents the contact area between the AM and electrolyte, τ denotes the time for applying the current pulse, ΔE_s denotes the steady-state voltage change, ΔE_t denotes the transient voltage change, and M_{AM} denotes the molecular weight of the AM; m_{AM} denotes the mass of the AM in the composite electrode, and V_M denotes the molar volume of the AM. Effective ionic and electronic conductivities of the composite electrode were evaluated via chronoamperometry (CA) using Solartron 1287 (Solartron Analytical). Effective ionic conductivity was measured using an electron-blocking cell with the following configuration: [Li metal | LPSCl | composite electrode | LPSCl | Li metal]. Effective electronic conductivity was measured using an ion-blocking cell with the following configuration: [SUS foil | composite electrode | SUS foil]. After applying a voltage of 0.1 V for 10 min, and the conductivity (σ) was calculated using Ohm's law ($R = V/I$) and Eq. 2.

$$\sigma = \frac{l}{S} \times \frac{1}{R} = \frac{l}{S} \times \frac{I}{V} \quad (2)$$

Here, I is the average of the last 100 measured current points, and V is the applied constant voltage. The resistance (R) was calculated using Ohm's law. l and S denote the thickness and area of the pellet, respectively.

2.4. Material characterization

Top-view and cross-sectional images of the samples were captured using field emission scanning electron microscopy (FE-SEM; Regulus 8230, Hitachi) equipped with energy-dispersive X-ray spectroscopy (Ultim Max, Oxford Instruments). Cross-sections of all samples were prepared using an ion milling system (ArBlade 5000, Hitachi). An air-protection holder was used to prevent air exposure. The surface morphologies of the current collector and primer layer were examined by atomic force microscopy (AFM; XE-100, Park Systems). Scans with a size of 10 $\mu\text{m} \times 10 \mu\text{m}^2$ were performed in non-contact mode using an NCHR cantilever. The three-dimensional (3D) AFM images were constructed

using XEI software. A surface and interfacial cutting analysis system (SAICAS; SAICAS-DN, Daipia Wintex Co., Ltd., Japan) was employed to measure the adhesive strength within the electrodes under an Ar atmosphere. The measurements utilized a 1-mm wide boron nitride blade at shear, rake, and clearance angles of 45°, 20°, and 10°, respectively. For peeling the primer layer from the current collector and the composite electrode from the current collector, the instrument operated in a constant load mode. A cutting load of 1.1–1.2 N was applied until the blade encountered an interface with significantly different physical properties. A peeling load of 0.8 N was then applied horizontally to detach the electrode or primer layer. Depth profiling of Cu C.C was performed using time-of-flight secondary ion mass spectrometry (ToF-SIMS; ToF-SIMS5, IONTOF). All analyses were performed with a primary Bi³⁺ beam (30 keV) over a 50 $\times$ 50 μm^2 raster area after sputtering with a secondary Cs⁺ beam (10 keV) over a 200 $\times$ 200 μm^2 raster area. The interface of composite and anode of Cu C.C, primer@Cu was analyzed by Focused Ion Beam-Scanning Electron Microscope (FIB-SEM; Helios NanoLab 600, FEI) and energy-dispersive X-ray spectroscopy (Ultim Max, Oxford Instruments). Cross-sectional electron beam absorbed current (EBAC) measurements were conducted using SEM (Teneo VolumeScope, FEI) equipped with EBAC instruments (miBot, Imina Technologies). The interfacial resistance between the electrodes and current collectors and the bulk resistivity of each electrode were measured using an electrode resistance meter (RM2610, Hioki). The measurements were performed over a the current range of 0–1 mA and voltage range of 0–0.5 V. Particle size analyses were performed by dispersing particles in toluene using Particle Size Analyzer (PSA; Microtrac S3500, Microtrac). Crystal structures of Cu C.C and primer@Cu were examined using grazing incidence X-ray diffraction (GIXRD). GIXRD patterns were measured using an X-ray diffractometer (D8 ADVANCE, Bruker) with monochromatic Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$) at a scanning speed of 2° min⁻¹ in the 2 θ range of 10°–90°, maintaining an incident angle of 2° or 5°. All film samples were mounted on a zero-background silicon plate within an air-tight holder to prevent exposure to air. The Young's moduli of the samples were calculated based on SEM images using SEM (Nova Nano SEM 200, FEI) equipped with nano-indentation instrument (Hysitron PI-85, Bruker). The tip was a cube corner, and the cross-sections of the specimens were measured. The loading force was 8 mN, with loading, holding, and unloading times of 4, 2, and 4 s, respectively. To examine the porosity of the electrode according to changes in binder content, analysis was conducted using a He gas pycnometer (BELPYCNO, MicrotracBEL). Since the pycnometer-based porosity measurement cannot measure closed pores, the true volume tends to be overestimated, meaning that it does not accurately reflect the actual porosity of the composite electrode [29]. However, this analysis was conducted to compare the pristine electrode with the electrode containing a primer layer, which has a reduced binder content. The results indicate that apart from a slight difference in porosity due to the binder content variation, no significant issues were observed. First, the composite electrode slurry casting that was previously conducted on the current collector was performed on the PET film, and then drying was conducted, and the composite electrode film was peeled off. A certain amount of the peeled-off electrode was placed into a 9 mm mold and pelletized under a pressure of 370 MPa. After removing the pellet from the mold, its height was measured to calculate the apparent volume. Subsequently, the pellet was placed in the pycnometer sample holder, and its true volume was measured using the pycnometer. Porosity was then calculated using Eq. 3.

$$\text{porosity} = \left(1 - \frac{V_{\text{true}}}{V_{\text{apparent}}} \right) \times 100 \quad (3)$$

2.5. Generation of 3D electrode structure

To characterize the electrode microstructure using a three-dimensional modeling and simulation approach, virtual electrode

structures were built using a stochastic generation method based on the volume fraction of each electrode component. The volume fraction was estimated from the electrode design parameters (composition, mass loading, and density) and the densities of individual components: NCM82 (4.70 g cm^{-3}), graphite (2.10 g cm^{-3}), LPSCl (1.88 g cm^{-3}), NBR (1.00 g cm^{-3}), SBS (1.00 g cm^{-3}), and super P (2.0 g cm^{-3}). The active material, solid electrolyte, and carbon binder domain were uniformly distributed in a 3D computational domain of $30 \times 30 \times 55 \mu\text{m}^3$ for the cathode and $50 \times 50 \times 65 \mu\text{m}^3$ for the anode, with a voxel size of $0.05 \mu\text{m}$, using GeoDict software (Math2Market, Germany). The thickness of the computational domain was smaller than the actual electrode thickness to save memory usage and reduce simulation time. The size distribution of each particle and the thickness of the primer layer were determined based on experimentally measured data (detailed modeling parameters are summarized in Table S1).

2.6. Electron/ion conduction simulation

Electron and ion conduction simulations were performed using the 3D electrode structures. The boundary conditions were imposed as $\Delta V = 1 \text{ V}$ under the Dirichlet condition along the thickness direction (z-direction), applying a constant potential to the designated planes (1 V at the $-z$ plane and 0 V at the $+z$ plane). Ohm's law was employed as the governing equation for both electron and ion conduction simulations. A finite volume method (FVM)-based partial differential equation solver was utilized to calculate the electronic and ionic current densities. Based on Ohm's law and the computed current densities, the electronic and ionic conductivities of the electrode were determined. Detailed input parameters are provided in Table S2[30–34].

3. Results and discussion

3.1. Physical analysis of primer layer

The primer solution comprised a carbon conductive agent and polymeric binder, with the weight ratio adjusted between 9:1 and 8:2 depending on the electrode type. Since the final electrode requires double-casting on the primer layer, it was essential to prevent the dissolution of the primer layer. Therefore, PVDF and NMP were used as the polymeric binder and solvent, respectively. Detailed information on the strategy for primer layers according to the choice of carbon conductive additives, ratio for polymeric binder can be found in the section, Materials and compositions suitable for primer layer in the [supplementary material \[35\]](#). SEM images of primer layers according to the morphology of each carbon conductive additive can be found in Figure S5. Adhesion force and bulk resistance according to variation in carbon conductive additive content in the primer layer can be seen in Figure S6. The carbon conductive agent and PVDF were uniformly dispersed using a horn sonicator. The prepared primer solution was then coated onto the current collector using the slurry-casting method and dried overnight in an oven. Detailed preparation methods are provided in the [supplementary material](#). Fig. 1a shows the top-view images of Al C.C and the primer@Al, whereas Fig. 1b shows corresponding top-view images of Cu C.C and the primer@Cu. These figures confirm uniform coating of the primer layer and increased surface roughness on both Al C.C and Cu C.C. Cross-sectional images of each primer (Figure S7) indicate a thickness of $2 \mu\text{m}$ for both primer@Al and primer@Cu, with noticeably rougher surfaces compared to the bare current collectors. The areal mass loading of primer@Al C.C is 0.29 mg/cm^2 and primer@Cu C.C is 0.27 mg/cm^2 . In general, an increased surface area improves the adhesive force at the interface; thus, the mechanical bonding force is

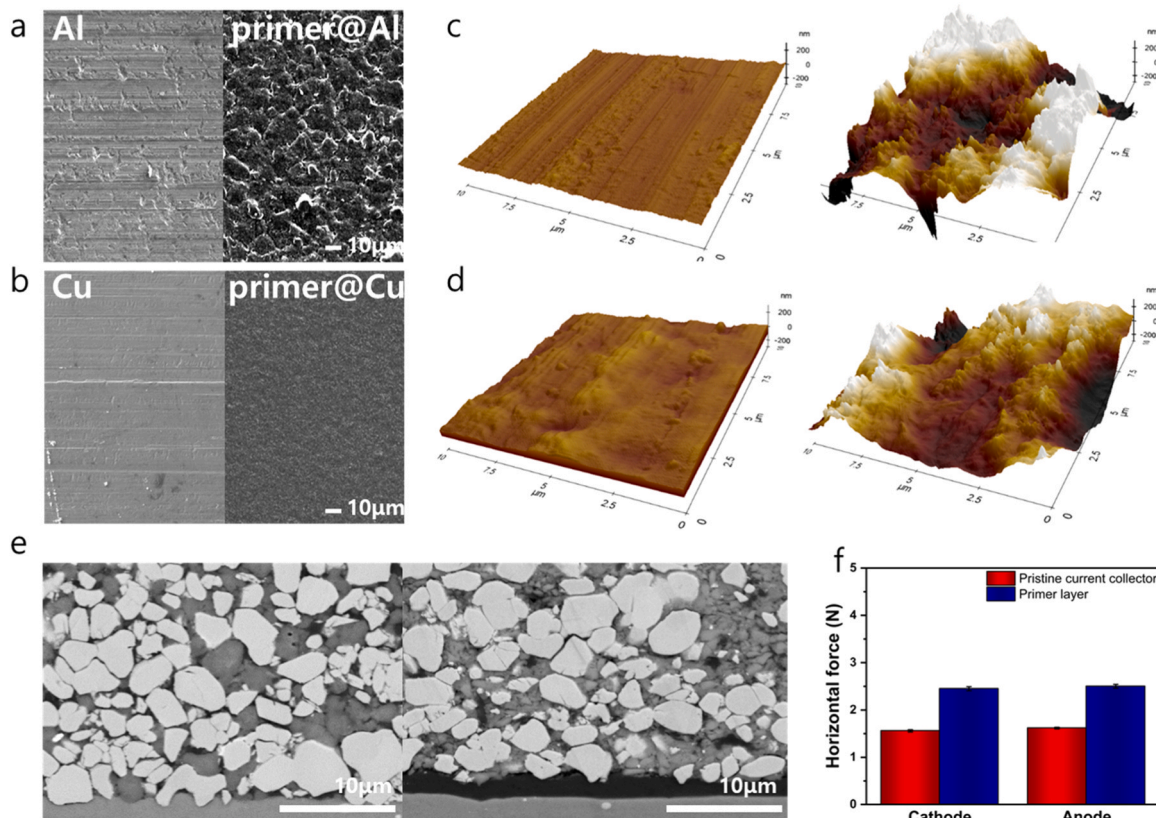


Fig. 1. Top-view SEM images of (a) Al C.C and primer@Al, and (b) Cu C.C and primer@Cu. 3D AFM images of (c) Al C.C (left) and primer@Al (right), and (d) Cu C.C (left) and primer@Cu (right). (e) Cross-sectional SEM images of electrodes on Al C.C (left) and primer@Al (right). (f) Adhesive force acting in the horizontal direction with current collector.

expected to be stronger with the primer coating[36].

AFM analysis further quantified the surface roughness and surface area. The 3D AFM images are shown in Fig. 1c and d. The surface roughness increased after primer coating. The corresponding roughness values for each sample are presented in Table S3. The R_q (root mean square roughness) and R_a (arithmetic average roughness) values of the primer-coated surfaces increased by approximately 20–30 times, and the surface area increased by approximately 5 %–15 %. Additional roughness data for the primer layers obtained through roughness line profiling are presented in Figure S8, further corroborating the increase in surface roughness. The interfacial adhesion force between each layer was quantitatively compared via SAICAS analysis. This method evaluates the adhesion force between the electrode and primer layers and the collectors. Since the metal current collectors exhibited significant differences in physical properties compared to the electrode and primer layers, the measurements were performed in constant load mode[37, 38]. The measurement results are presented in Fig. 1f. For primer@Al, the interfacial adhesion force was 2.45 N, representing a 57 % increase compared to Al C.C (1.56 N). For primer@Cu, the interfacial adhesion force was 2.50 N, representing a 54 % increase compared to Cu C.C (1.62 N). Considering the adhesion force generated by the lamination of electrode particles onto the metal current collector, the addition of a primer layer enhances the interfacial adhesion force.

3.2. Electrochemical characterization of primer layer

CV was employed to evaluate the electrochemical stabilities of LPSCl and the current collector interfaces. Measurements were conducted in potential windows of 2.7–4.3 V (vs. Li^+/Li) for Al C.C and 0.01–1.5 V (vs. Li^+/Li) for Cu C.C, at a rate of 0.1 mV/s. The primer@Al electrode exhibited no significant side reactions compared to Al C.C, with a similar electrochemical reaction as that of LPSCl (Fig. 2a). Side reactions between Cu and SSEs due to the sulfur-philic properties of Cu have been previously highlighted[39,40]. Hydrogen sulfide (H_2S), a toxic gas generated when SSE comes into contact with moisture (H_2O), is a major factor in Cu corrosion[41]. The cell was fabricated in an Ar-filled glove box, ensuring that CV measurements were conducted under conditions with significantly suppressed H_2O . This allowed the electrochemical reactivity to be confirmed only when Cu C.C and LPSCl were in direct contact. A current peak attributed to the redox reaction of LPSCl in Cu C.C was observed as expected, with the redox peak intensity increasing with each cycle. This indicates that when Cu C.C is in direct contact with the SSE, an electrochemical side reaction occurs, accelerating the electrolyte oxidation. In contrast, for primer@Cu, no peak associated with LPSCl oxidation was detected, confirming that the primer layer effectively suppresses such side reactions or SSE oxidation (Fig. 2b). Moreover, considering the application to a full cell, performing CV over a

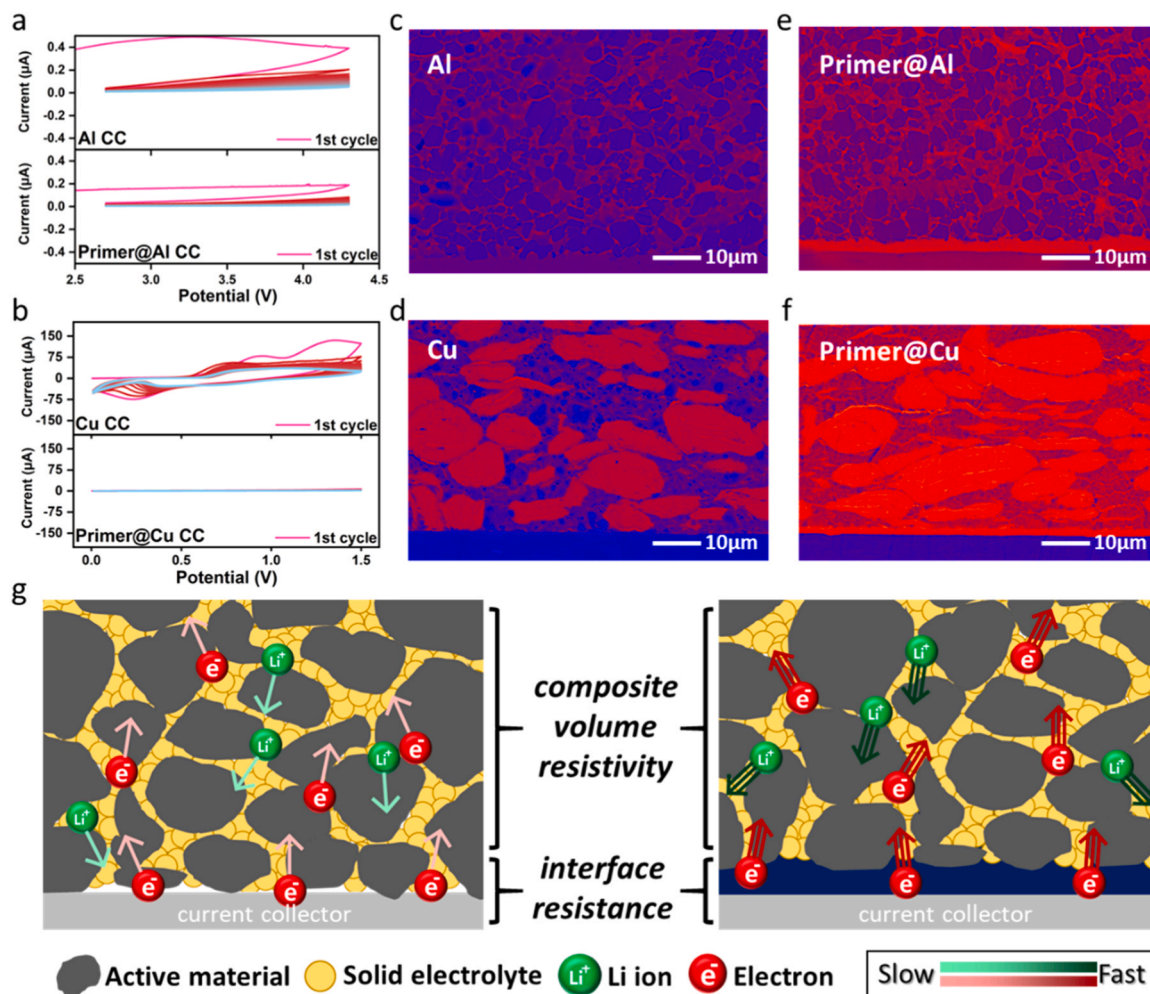


Fig. 2. CV profiles of symmetric cells using (a) bare Al C.C and primer@Al, and (b) bare Cu C.C and primer@Cu. Pink represents the first CV cycle and gradually changes to light blue as CV cycling progresses. All symmetric cells were operated at 30°C without any additional external pressure. CV measurements were performed in the voltage range of 0.01–1.5 V (vs. Li^+/Li). Electrical resistance images based on EBAC measurements: (c) Al C.C, (d) Cu C.C, (e) primer@Al, and (f) primer@Cu. (g) Schematic of Li-ion and electron movement within composite electrode with current collector. The locations measured with an electrode resistance meter are described. The blue interlayer on the right electrode in (g) represents the primer layer.

wide voltage range shows that Cu C.C undergoes more severe side reactions. In contrast, when the primer layer is applied, it remains relatively stable even within the expanded voltage range (Figure S9). The AFM and SAICAS results confirmed enhanced bonding area and binding strength. The polymeric binder—an electrochemically inactive material with electrically insulating properties—introduces resistance to Li-ion and electron transport at the electrolyte–AM interface. Therefore, the reduction of the polymeric binder can be expected to improve electronic and ionic conductivities; therefore, the binder content was minimized while ensuring that the large-area electrode for the pouch-type cell maintained its structural integrity (Figure S10). Consequently, we performed electrochemical evaluations using electrodes with different binder contents depending on the presence or absence of primer layer. For the cathode, a slurry with 1.5 wt% NBR content was cast for Al C.C, whereas a slurry with 0.2 wt% NBR content was cast for primer@Al. For the anode, Cu C.C was cast in a slurry with a SBS content of 3 wt%, and primer@Cu was cast in a slurry with a SBS content of 2 wt%. The ratio of the detailed electrode composition is described in the experimental section. The varying electrode compositions are expressed as LBC and high binder content (HBC). Fig. 2c–f present the EBAC images of the four electrodes: HBC/Al C.C, LBC/primer@Al, HBC/Cu C.C, and LBC/primer@Cu. The difference in electronic conductivity of each cast electrode is confirmed through EBAC analysis. The higher the electronic conductivity, the stronger the red intensity [42]. LBC/primer@Al and LBC/primer@Cu exhibit stronger red intensity compared to HCB/Al C.C and HBC/Cu C.C at both cathode and anode, indicating higher electronic conductivity even at identical conductive material contents. In addition, to compare the interfacial affinity of the primer layer and the quantitative electronic conductivity within the electrode, the interfacial and composite volume resistances were measured using an electrode resistance meter. As shown in Fig. 2g, the interfacial resistance indicates electron transfer ease from the current collector to the composite electrode. Also, the composite volume resistance represents the resistance to

the electron movement within the composite electrode. The primer layer exists between the electrode and the current collector, which directly improves the contact area and reduces the interfacial resistance. In the composite electrode with a reduced amount of the electrochemically inactive binder, the composite volume resistance is also decreased. The results are presented in Table S4 [43]. The primer layers reduced the interfacial resistances of the cathode and anode by 91.4 % and 88.8 %, respectively, and the composite volume resistances by 72.5 % and 70.8 %, respectively. These results show that the primer layer improves the electronic conductivity within the electrode by reducing the interfacial resistance and optimizing the electrode composition.

3.3. Electrochemical performance of half cell

Half-cell tests were conducted to assess the effect of primer layer on electrochemical performance. For the cathode, the composite cathode slurry was cast onto Al C.C and primer@Al at the optimized ratio above. The half-cell test was performed within a voltage window of 2.1–3.7 V ($\text{Li}^+/\text{Li-In}$), with the first cycle serving as a formation cycle at 0.05 C ($1 \text{ C} = 180 \text{ mA g}^{-1}$). Original Al C.C electrode exhibited a specific capacity of 191.0 mAh g^{-1} , whereas the optimized primer@Al electrode with a reduced NBR content exhibited a capacity of 202.2 mAh g^{-1} , indicating an increased initial capacity (Fig. 3a). In addition, the optimized primer@Al system exhibited 86.7 % capacity retention after 50 cycles (0.1 C) despite a greater volume change owing to the high reversible capacity (Fig. 3b). This is higher than the 73.0 % capacity retention of Al C.C [44,45]. The increased discharge capacity and improved cycle performance can be primarily attributed to the introduction of primer layer and reduction of electrochemically inactive binder content. It is assumed that reducing the binder content increases the contact area between the cathode AM and SSE [27].

To verify the increase in contact area, GITT analysis was performed. The results confirmed that the AM coverage increased from 27.8 % for

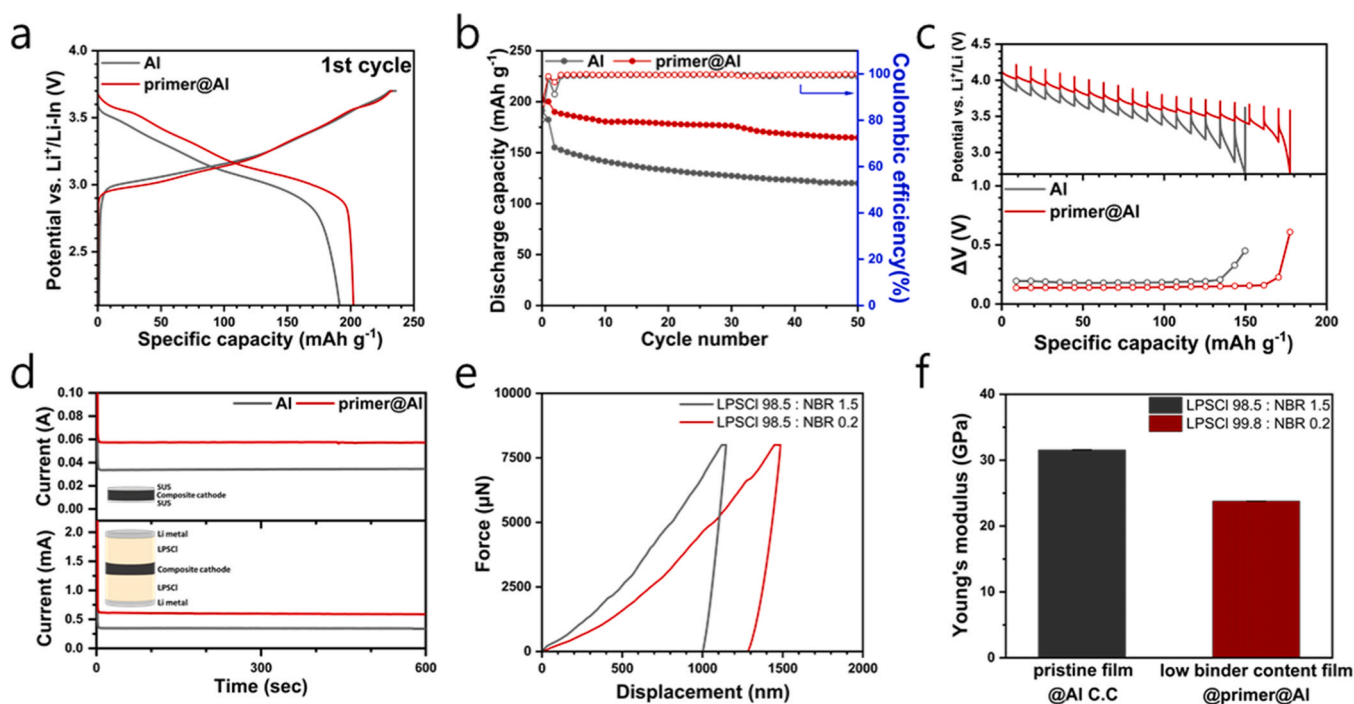


Fig. 3. Electrochemical characterization of half cells using pristine and primer-coated current collectors at 30 °C. (a) First-cycle charge–discharge voltage profiles of Al C.C and primer@Al, and (b) their corresponding cycling performance. (c) GITT curves and polarization plots derived from GITT results. Half-cell test and GITT measurements were conducted at 30 °C without any additional external pressure and in the voltage range of 2.1–3.7 V (vs. $\text{Li}^+/\text{Li-In}$). (d) CA measurements and configuration of ion-blocking cells (up) and electron-blocking cells (down). CA measurements were performed at room temperature under an external pressure of 10 MPa with a voltage of 0.1 V. (e) Representative nano-indentation analysis curves for each sample. (f) Columns represent Young's modulus, and error bars represent standard errors.

HBC/Al C.C to 62.7 % for LBC/primer@Al (Fig. 3c), suggesting higher effective ionic conductivity in the composite electrode due to improved Li^+ conduction pathways [16,46]. To measure the Li^+ diffusion coefficient of each active material, GITT measurements and BET analysis were first conducted using a liquid electrolyte, and the results are shown in Figure S11 and S12. To confirm improved Li^+ conduction pathway, electrochemical analysis using CA was performed, showing LBC/primer@Al exhibited a higher effective ionic conductivity of 0.10 mS cm^{-1} compared to 0.05 mS cm^{-1} of HBC/Al C.C (Fig. 3d) [47]. In addition, the electronic conductivity increased from 5.44 to 9.29 mS cm^{-1} . This trend aligns with the EBAC analysis (Fig. 2c–f) and electrode resistance results (Table S4), demonstrating the balanced ionic conductivity and increased electronic conductivity of the composite cathode. The configuration of the pellet used for each CA measurement is shown in Fig. 3d. An electron-blocking cell was prepared for ionic conductivity measurements, and an ion-blocking cell was prepared for electronic conductivity measurements. Similar to the cathode half cells, anode half cells with Cu C.C and primer@Cu were also evaluated for electrochemical performance in the voltage range of 0.01 – 1.5 V (Li^+/Li) (Figure S13a and b), with the first cycle serving as the formation cycle at 0.05 C ($1 \text{ C} = 372 \text{ mA g}^{-1}$). The primer@Cu electrode exhibited an initial capacity of 361.8 mAh g^{-1} , higher than 341.0 mAh g^{-1} for Cu C.C. In addition, primer@Cu exhibited 86.3% capacity retention after 50 cycles (0.1 C) compared to 79.3% for Cu C.C. Similar to the cathode, the GITT results of primer@Cu reveal a larger contact area (62.0%) between the AM and SSE compared to Cu C.C (47.8%) (Figure S13c). In addition, corresponding ionic and electronic conductivities for primer@Cu were significantly higher (0.10 and $118.98 \text{ mS cm}^{-1}$, respectively) than for Cu C.C (0.08 and 55.30 mS cm^{-1} , respectively) (Figure S13d). These results demonstrate that improving the interfacial adhesion between the electrode and current collector and controlling the electrochemically inactive material content (binder content) are effective strategies for excellent cell performance. Reducing the binder content also affects mechanical properties of the electrode. In contrast to the relatively hard property of SSE, the polymer binder has relatively elastic properties. Therefore, it was expected that the electrode would become slightly harder when the binder content was reduced, but the actual analysis reveals that reducing the binder content decreases the Young's Modulus of the electrode (Fig. 3e and f), with values of 31.6 GPa for pristine film@Al C.C and 23.8 GPa for LBC film@primer@Al. To increase the reliability of the nano-indentation analysis, simpler electrodes containing only LPSCl and NBR, rather than the composite electrode, were used [48–50]. The pristine film comprised of 98.5% LPSCl and 1.5% NBR, and the LBC film comprised of 98.5% LPSCl and 0.2% NBR. While polymeric binders contribute to the cycle performance, they also deteriorate the electrochemical performance and energy density [51–53]. Therefore, minimizing binder content is crucial, especially for slurry-casting methods. In addition, many studies have demonstrated that fluctuating elastic modulus is not an appropriate indicator of electrochemical performance [48,54,55]. Overall, achieving appropriate binder content and elastic modulus is crucial, and they should be optimized through electrochemical evaluation. Although further studies are needed, our findings suggest that reducing binder content and elastic modulus in the primer system does not compromise mechanical integrity. In fact, we hypothesize that reduced binder content may help alleviate the stress induced by repeated lithiation and delithiation cycles or, at the very least, does not have a fatal effect on the macroscopic cracks of the composite electrode (Figure S14). Finally, He gas Pycnometry was conducted to observe the changes in porosity depending on the binder content. The analysis results are presented in Table S5. Although porosity analysis using He gas pycnometer is not highly precise due to errors caused by closed pores, it was performed for relative comparison between electrodes. A pellet containing the active material, LPSCl, carbon conductive additive, and binder was prepared for the analysis, and the detailed pellet fabrication process can be found in the material characterization. The analysis results indicate that the

difference in porosity depending on the binder content is minimal. It was expected that Li-ion conduction issues caused by porosity differences would not be significant, confirming that only the binder content, which acts as an insulator within the electrode, is crucial. As previously noted, we believe that the development to continuously reduce the binder content is crucial for improving the electrochemical performance of large-scale ASSBs [52].

3.4. 3D electrode modeling & conduction simulation

To investigate the effects of introducing the primer system into electrodes and the changes in ionic and electronic conductivity resulting from reduced binder content in composite electrodes, we conducted digital twin simulations. We analyzed the impact of the primer system and binder content on both cathode and anode, confirming the mechanisms behind the improvements in electrochemical performance. The results for the cathode are presented in the main text, while the results for the anode are provided in the supplementary material. The 3D model of the designed composite cathode electrode is shown in Fig. 4a. In other words, we confirmed that the best electrochemical properties were achieved when the primer layer was introduced, and the binder content was reduced. We believe this is because the primer layer primarily enhances electronic conductivity, while the reduction in binder content mainly improves ionic conductivity. To validate this, we designed three different models like; 1. A high binder content electrode on a pristine current collector (HBC), 2. A high binder content electrode on a primer layer (HBC/primer), and 3. A low binder content electrode on a primer layer (LBC/primer). The electrode components include active material, S-LPSCl, and a carbon-binder composite domain (CBD) consisting of Super P and binder. Since Super P has a significantly smaller volume compared to the active material and S-LPSCl, it was modeled as a composite with the binder. Fig. 4b presents a 2D cross-sectional image of the interface between the composite electrode layer and the current collector. When the primer layer is present, the active material penetrates deeper into the Al current collector, which was simulated using particle stochastic generation. Using the 3D model, we conducted electronic and ionic conductivity simulations based on Ohm's Law. The applied boundary conditions were set to 1 – 0 V (Fig. 4c). The mapping of the electron current density obtained from the simulations is shown in Fig. 4d. When the primer layer is absent, the contact area between the composite electrode layer and the current collector is relatively small, leading to current crowding at localized contact points. This current concentration contributes to an increase in resistance, as evidenced by the potential distribution within the electrode (Fig. 4e). As shown in Fig. 4e, in the HBC/Al C.C. system without a primer layer, the potential drops sharply near the interface with the current collector (normalized thickness = 0) due to high resistance. These results indicate that the improvement in electronic conductivity resulting from the primer layer is attributed to the mitigation of current crowding at the interface, and that binder content reduction further enhances electronic conductivity. Next, the mapping of ionic current density is shown in Fig. 4f. While the primer layer does not significantly enhance ionic conductivity, reducing binder content results in more intimate interfacial contact between S-LPSCl particles, creating more ionic conduction pathways that were previously blocked. Consequently, the ionic current density increases. The diversification of ionic conduction pathways is illustrated in Fig. 4g, confirming the improved ionic conductivity in LBC/Primer@Al. The 3D modeling and simulation results for the anode are presented in Figure S15. Similar to the cathode, the application of the primer layer and reduction of binder content improved both electronic and ionic conductivity. However, in the anode, the enhancement in electronic conductivity due to the primer layer was more pronounced than in the cathode. This can be attributed to the intrinsic properties of graphite, which has a larger particle size and higher electronic conductivity compared to NCM82, making the contact improvement effect of the primer layer more significant. To further understand and visualize the

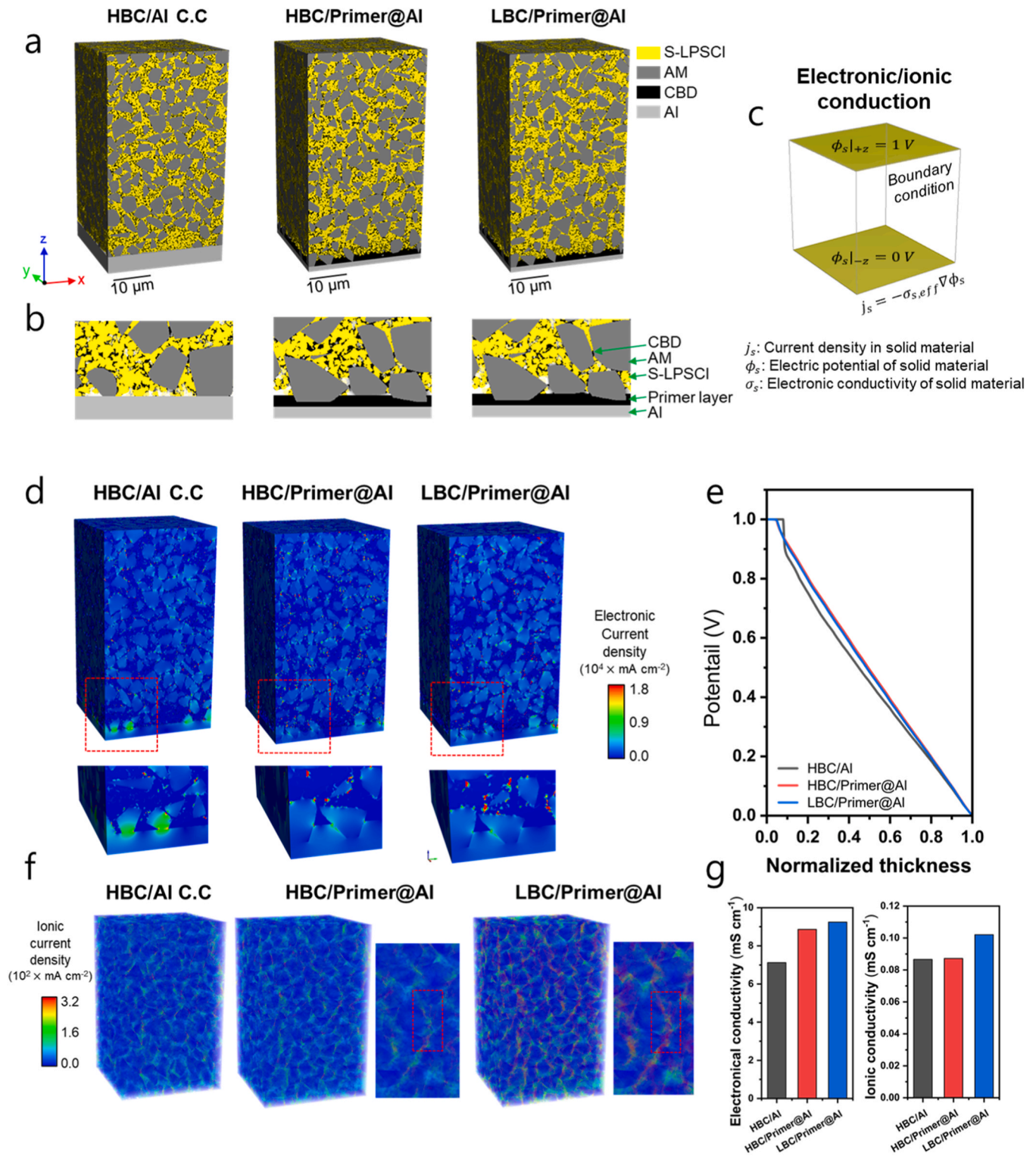


Fig. 4. 3D Modeling and simulation approach for electrode microstructure characterization. (a) 3D electrode model of composite cathode (HBC/Al, HBC/primer@Al, and LBC/primer@Al). (b) 2D cross-sectional images of 3D cathode model. (c) Governing equation and boundary condition to characterize electronic and ionic conductivity of 3D cathode model. (d) electronic current density 3D/2D color map of the composite cathode. (e) Electronic conductivity of the cathode estimated from the 3D model-based electron conduction simulation. (f) Ionic current density 3D/2D color map of the composite cathode. (g) Electronic, ionic conductivity of the cathode estimated from the 3D model-based ion conduction simulation.

impact of primer layer introduction and binder content reduction on electrochemical performance, 3D modeling and simulations were conducted. While the morphological relationships among electrode components (e.g., active material shape, primer layer induced contact changes, concave binder structures) were incorporated as accurately as

possible, some simplifications of complex electrode structures were inevitable. However, the enhanced contact area between the composite electrode layer and the current collector due to the primer layer was clearly visualized, along with the diversification of ionic conduction pathways resulting from binder content adjustment. Ultimately, these

findings elucidate the fundamental mechanisms behind the improvement in battery performance through enhanced effective electronic and ionic conductivity

3.5. Primer layer as passivation layer

Fig. 5 highlights the primer layer's effectiveness in preventing corrosion. Traditionally, Cu C.C has been used as an anode current collector due to its low cost and ease of processing; however, its susceptibility to corrosion due to reactions with sulfide poses a major limitation [56].

In this study, two well-established reaction mechanisms between Cu and sulfides were analyzed to observe the effect of primer layer, and analyzed the passivation effect. The first reaction is a corrosion process triggered by H_2S generation from the reaction between air moisture and sulfide—a well-known process[41]. The second reaction involves corrosion caused by an electrochemical reaction resulting from direct contact with an SSE[57]. The primer layer effectively prevents corrosion in both cases. Fig. 5a–c presents the GIXRD results for Cu C.C before corrosion, Cu C.C and primer@Cu exposed to H_2S , and Cu C.C and primer@Cu after galvanostatic cycling, respectively. There may be differences in the intensity due to the preferential orientation of the specimen surface[58]. Therefore, we focused on identifying the positions of the side-reaction peaks rather than performing accurate phase analysis through library matching. When Cu C.C was exposed to H_2S , multiple side-reaction peaks appeared except those associated with Cu, indicating significant corrosion. In contrast, primer@Cu exhibited no peaks other than those associated with Cu. In addition, after galvanostatic cycling, we confirmed that the side reactions were suppressed in primer@Cu, whereas Cu C.C showed corrosion due to the electrochemical side reactions. These results align with prior studies and demonstrate the primer layer's role as a passivation barrier on the current collector, preventing Cu corrosion from both H_2S and

SSE-induced reactions[59]. Additional analyses were conducted to more clearly determine whether Cu C.C corrodes before and after cycling and, if so, whether the primer layer can prevent corrosion. FIB was used to perform cross-sectional milling and line profiling on the electrodes before and after cycling. In the case of LBC/primer@Cu, a $2\mu m$ thick primer layer was observed on the surface both before and after cycling (Figure S16a and b). Before cycling, Cu corrosion at the surface was negligible and undetectable. After cycling, Sulfur was detected within the primer layer, but it did not penetrate through the layer to reach the Cu C.C. In contrast, for Cu C.C, Cu corrosion was observed on the surface immediately after electrode fabrication using the sulfide solid electrolyte, even before cycling. After cycling, this corrosion effect became more pronounced, progressively penetrating deeper into the Cu layer (Figure S16c and d). The analysis was conducted after 50 cycles, and while variations may exist depending on the specific cell, the corrosion depth increased from $0.2\mu m$ before cycling to $0.4\mu m$ after cycling, doubling in depth. These results demonstrate that the introduction of the primer layer not only enhances electrochemical performance by facilitating uniform current distribution but also mitigates current collector damage caused by direct contact with the sulfide solid electrolyte and associated electrochemical side reactions. Fig. 5d–g shows the ToF-SIMS results to observe the Cu/S side reactions before and after pouch-type full-cell fabrication and cycling. The fabrication processes of all electrodes were performed in an Ar-filled glove box, and the subsequent electrode cutting, tap welding, and sealing were performed in a dry room. Fig. 5d and e show electrodes being exposed to air containing moisture for approximately 60 min prior to sealing. Fig. 5f and g show the electrodes obtained from a pouch-type full cell operated for an additional 100 cycles. The ToF-SIMS results confirm the reaction between Cu and S on the Cu C.C surface, consistent with the GIXRD results and unrelated to the electrochemical tests (Fig. 5d). In addition, the surface reactions between Cu and S intensified after cycling (Fig. 5f). Consistent with the GIXRD results, the primer layer significantly

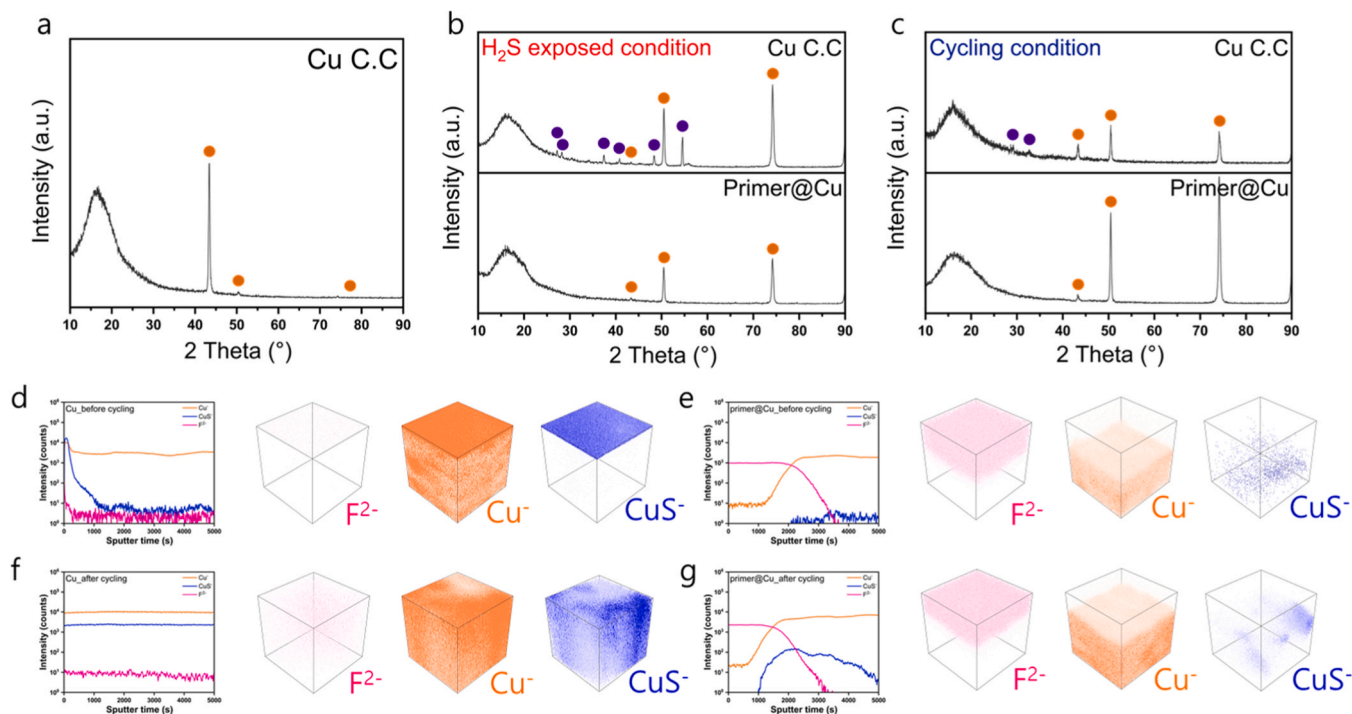


Fig. 5. GIXRD patterns of (a) pristine Cu C.C without any chemical or electrochemical corrosion. (b) Cu C.C (up) and primer@Cu (down) exposed to H_2S gas in a dry room. (c) Cu C.C (up) and primer@Cu (down) after galvanostatic cycling. The orange circles in the GIXRD graph represent Cu metal, and the purple circles represent the by-products of other side reactions. ToF-SIMS depth profiles and 3D reconstruction images of F^{2-} , Cu^- , and CuS^- corresponding to (d) Cu C.C before cycling, (e) primer@Cu before cycling, (f) Cu C.C after cycling, and (g) primer@Cu after cycling. An additional benefit of the primer layer, particularly on the Cu substrate, is its role as a passivation layer. The primer layer on Cu C.C prevents side reactions between Cu and sulfides, maintaining the electronic conductivity of the current collector and ensuring strong adhesion force to the electrode, thereby improving cyclability.

suppressed Cu corrosion caused by H_2S (Fig. 5e), confirming its ability to physically block H_2S gas and mitigate corrosion. In addition, the reaction between Cu and S was significantly alleviated after cycling, emphasizing the importance of preventing the initial side reactions between Cu and S (Fig. 5g). Therefore, the primer layer can suppress side reactions on the Cu surface in a dry room environment, even after long-term cycling, thereby maintaining the electronic conductivity while preventing degradation of adhesion with the electrode. In the commercial manufacture of sulfide-based ASSBs, where slurry casting and drying of the electrode are conducted in dry room environments, side reactions between Cu and S pose significant challenges. To address these challenges, passivation layers, like the proposed primer layer, will be crucial.

3.6. Pouch-type full-cell test

We fabricated a pouch-type full cell by coupling the electrodes previously evaluated as half cells. The results are presented in Fig. 6. The full cell combining primer@Al and primer@Cu (primer full cell) exhibited a discharge capacity of 201.7 mAh g^{-1} in the first cycle, which was approximately 10 % higher than that of the full cell without primers (uncoated full cell, 182.0 mAh g^{-1}) (Fig. 6a). Additionally, the initial Coulombic efficiency was significantly improved, increasing from 75.6 % to 80.3 %. Furthermore, under room temperature conditions without external pressure, the primer full cell exhibited 80.7 % capacity retention after 100 cycles. In contrast, the uncoated full cell showed only 64.8 % capacity retention (Fig. 6b). The superior discharge capacity and

cyclability is attributed to the primer layer's ability to effectively eliminate insulating regions within the composite electrode, diversifying the movement pathways for Li ions and electrons. High adhesion forces between the current collector and electrode is another contributing factor. This allowed for stable cycle retention even with severe volume changes associated with high reversible capacity (Figure S14).

To investigate the impact of high-electronic-conductive graphite on full cell operation, ToF-SIMS for image mapping was performed. The results of this image mapping are presented in Figure S17. Given that graphite has significantly higher electronic conductivity compared to oxide-based cathode materials such as NCM, it can accelerate the electrochemical decomposition of LPSCl. Therefore, an evaluation of the interface behavior between graphite and LPSCl was conducted. The results revealed that Li_3P , Li_2S , and LiCl were mainly formed as the anodic solid electrolyte interphase (SEI) products, which are known to exhibit relatively low impedance values[60]. This indicates that, compared to P_2S_5 , which forms as the cathodic solid electrolyte interphase at high voltage and has relatively high impedance values, the impedance growing caused by using graphite in the full cell was not significant. Fig. 6c and S18 shows the rate performance of each full cell. For the primer full cell, high-rate performance was dramatically improved despite the cell exhibiting the same ion conductivity as the SSE membrane. In particular, at 1 C rate, the primer full cell achieved a reversible capacity of 46.7 mAh g^{-1} , which is significantly higher than the 2.26 mAh g^{-1} reversible capacity exhibited by the uncoated full cell. These results indicate that the optimization of electrode composition is important for improving battery performance. Fig. 6d compares the

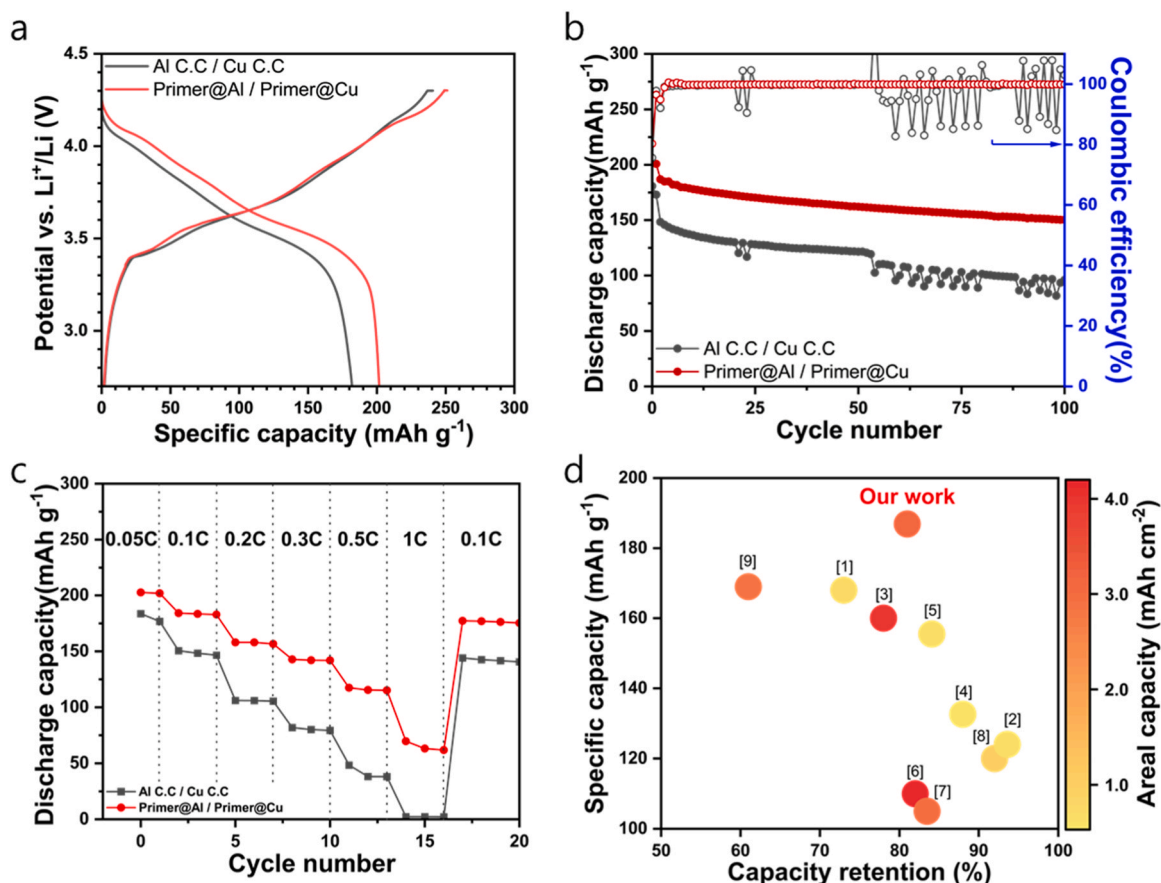


Fig. 6. Electrochemical performance of $40 \times 30 \text{ mm}^2$ pouch-type full cells with primer-coated and uncoated electrodes. (a) First charge-discharge cycle voltage profiles. (b) Corresponding cycling performance. Both cells with primer-coated and bare electrodes were operated at 0.1 C after a formation cycle of 0.05 C. (c) Rate performance of each pouch-type full cell. All pouch-type full cell tests were conducted at 30°C without any additional external pressure and in the voltage range of 2.7–4.3 V (vs. Li^+/Li). (d) Comparison of pouch-type ASSBs reported in the literature with those presented in our work. The references used for comparison are listed in Table S6.

specific capacity, cyclability, and areal capacity of the primer-applied full cell with those reported in previous studies (Table S6) [48,52, 61–67]. Although there are differences in the battery operating conditions, the full cell in the present study achieved an areal capacity of 3.4 mAh cm⁻² (0.05 C), 3.1 mAh cm⁻² (0.1 C) while exhibiting excellent specific capacity and cyclability. These findings are more relevant to the commercialization of ASSBs as they were achieved under normal pressure and temperature conditions.

4. Conclusion

This study demonstrated the significant improvement in the electrochemical performance of a pouch-type full cell by introducing a primer layer and optimizing the electrode configuration. First, the primer layer effectively resolved the issue of insufficient adhesion force between the electrode and current collector due to the limited selection of organic solvents and polymeric binders. Subsequently, the improved interfacial adhesion force allowed for a reduction in the content of the insulating binder in the electrode, increasing the contact area between the AM and SSE in the composite electrode, as well as between the SSE particles. The increased contact area diversified Li-ion pathways, thereby increasing the effective ionic and electronic conductivities of the composite electrode. The reduced binder content in the electrode lowered the Young's Modulus but did not cause mechanical damage to the electrode; instead, it improved the electrochemical performance. Based on these results, polymeric binders such as NBR and SBS, without special functional groups, were found to be necessary only during the cell fabrication process, and their presence in the electrode was determined to have no positive effect on the electrochemical performance of ASSBs. If a fabrication process capable of uniformly applying sufficient pressure such as WIP is prepared, the reduced binder content has no adverse effect on the long cycling stability. In addition, the improved contact area between the activated active material and the solid electrolyte due to the reduced binder content showed a positive effect on not only the discharge capacity but also the cyclability. The primer layer prevents electrode delamination, ensuring long-term operation of the pouch-type full cell even under no external pressure. It also improves the cycle life performance by suppressing the chemical and electrochemical side reactions between Cu and sulfides. The proposed primer system provides a practical solution for the fabrication of large-scale commercial sulfide-based ASSBs, particularly those fabricated in dry room conditions. In this study, we propose that the current collector design and electrode composition is important for the practical performance of pouch-type full cells at normal pressures and temperatures, unlike the laboratory-scale small-sized cells that operate under high stack pressures and elevated temperatures.

CRedit authorship contribution statement

Oh Hyeonseong: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kim Jun Tae:** Formal analysis, Data curation. **Shin Hyeon-Ji:** Writing – review & editing, Formal analysis. **Kim A-Yeon:** Formal analysis. **Bak Cheol:** Formal analysis. **Kim Sang-Ok:** Funding acquisition. **Chung Kyung Yoon:** Funding acquisition. **Mun Junyoung:** Resources. **Kim Jongsoo:** Resources. **Lee Yong Min:** Data curation, Resources. **Lee Sang-Young:** Resources, Supervision. **Jung Hun-Gi:** Writing – review & editing, Supervision, Resources, Project administration..

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Korea Institute of Science and Technology (KIST) institutional program (2E32581); Development Program of Core Industrial Technology funded by the Ministry of Trade, Industry & Energy of Korea (No. 20007045); National Research Council of Science & Technology (NST) grant and the National Research Foundation from Ministry of Science and ICT of Korea (No. GTL24011-000 and No. NRF-2022M3J1A1054151, respectively).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.mser.2025.100970.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

- [1] D. Larcher, J.-M. Tarascon, Towards greener and more sustainable batteries for electrical energy storage, *Nat. Chem.* 7 (1) (2015) 19–29.
- [2] N. Kittner, F. Lill, D.M. Kammen, Energy storage deployment and innovation for the clean energy transition, *Nat. Energy* 2 (9) (2017) 1–6.
- [3] Q. Wang, P. Ping, X. Zhao, G. Chu, J. Sun, C. Chen, Thermal runaway caused fire and explosion of lithium ion battery, *J. Power Sources* 208 (2012) 210–224.
- [4] G.G. Eshetu, J.-P. Bertrand, A. Lecocq, S. Grugeon, S. Laruelle, M. Armand, G. Marlair, Fire behavior of carbonates-based electrolytes used in Li-ion rechargeable batteries with a focus on the role of the LiPF₆ and LiFSI salts, *J. Power Sources* 269 (2014) 804–811.
- [5] Z. Gao, H. Sun, L. Fu, F. Ye, Y. Zhang, W. Luo, Y. Huang, Promises, challenges, and recent progress of inorganic solid-state electrolytes for all-solid-state lithium batteries, *Adv. Mater.* 30 (17) (2018) 1705702.
- [6] G. Xi, M. Xiao, S. Wang, D. Han, Y. Li, Y. Meng, Polymer-based solid electrolytes: material selection, design, and application, *Adv. Funct. Mater.* 31 (9) (2021) 2007598.
- [7] K.-H. Park, K. Kaup, A. Assoud, Q. Zhang, X. Wu, L.F. Nazar, High-voltage superionic halide solid electrolytes for all-solid-state Li-ion batteries, *ACS Energy Lett.* 5 (2) (2020) 533–539.
- [8] S. Chen, D. Xie, G. Liu, J.P. Mwizerwa, Q. Zhang, Y. Zhao, X. Xu, X. Yao, Sulfide solid electrolytes for all-solid-state lithium batteries: Structure, conductivity, stability and application, *Energy Storage Mater.* 14 (2018) 58–74.
- [9] J. Janek, W.G. Zeier, A solid future for battery development, *Nat. Energy* 1 (9) (2016) 1–4.
- [10] K.J. Kim, M. Balaish, M. Wadaguchi, L. Kong, J.L. Rupp, Solid-state Li–metal batteries: challenges and horizons of oxide and sulfide solid electrolytes and their interfaces, *Adv. Energy Mater.* 11 (1) (2021) 2002689.
- [11] A. Miura, N.C. Rosero-Navarro, A. Sakuda, K. Tadanaga, N.H. Phuc, A. Matsuda, N. Machida, A. Hayashi, M. Tatsumisago, Liquid-phase syntheses of sulfide electrolytes for all-solid-state lithium battery, *Nat. Rev. Chem.* 3 (3) (2019) 189–198.
- [12] P. Adeli, J.D. Bazak, K.H. Park, I. Kochetkov, A. Huq, G.R. Goward, L.F. Nazar, Boosting solid-state diffusivity and conductivity in lithium superionic argyrodites by halide substitution, *Angew. Chem. Int. Ed.* 58 (26) (2019) 8681–8686.
- [13] J. Wu, S. Liu, F. Han, X. Yao, C. Wang, Lithium/sulfide all-solid-state batteries using sulfide electrolytes, *Adv. Mater.* 33 (6) (2021) 2000751.
- [14] J. Wu, L. Shen, Z. Zhang, G. Liu, Z. Wang, D. Zhou, H. Wan, X. Xu, X. Yao, All-solid-state lithium batteries with sulfide electrolytes and oxide cathodes, *Electrochem. Energy Rev.* 4 (2021) 101–135.
- [15] P. Lu, Y. Wu, D. Wu, F. Song, T. Ma, W. Yan, X. Zhu, F. Guo, J. Lu, J. Peng, Rate-limiting mechanism of all-solid-state battery unravelled by low-temperature test-analysis flow, *Energy Storage Mater.* 67 (2024) 103316.
- [16] J.T. Kim, H.-J. Shin, A.-Y. Kim, H. Oh, H. Kim, S. Yu, H. Kim, K.Y. Chung, J. Kim, Y.-K. Sun, An argyrodite sulfide coated NCM cathode for improved interfacial contact in normal-pressure operational all-solid-state batteries, *J. Mater. Chem. A* 11 (38) (2023) 20549–20558.
- [17] A.-Y. Kim, F. Strauss, T. Bartsch, J.H. Teo, T. Hatsukade, A. Mazilkin, Janek, J. r.; Hartmann, P.; Brezesinski, T. Stabilizing effect of a hybrid surface coating on a Ni-rich NCM cathode material in all-solid-state batteries, *Chem. Mater.* 31 (23) (2019) 9664–9672.
- [18] L. Yong-Gun, F. Satoshi, J. Changhoon, N. Suzuki, Y. Nobuyoshi, O. Ryo, S. Tomoyuki, R. Saebom, J.H. Ku, W. Taku, High-energy long-cycling all-solid-state lithium metal batteries enabled by silver–carbon composite anodes, *Nat. Energy* 5 (4) (2020) 299–308.
- [19] X. Zhao, P. Xiang, J. Wu, Z. Liu, L. Shen, G. Liu, Z. Tian, L. Chen, X. Yao, Toluene tolerated Li₉ 88GeP₁ 96Sb₀ 04S₁₁ 88Cl₀ 12 solid electrolyte toward ultrathin membranes for all-solid-state lithium batteries, *Nano Lett.* 23 (1) (2022) 227–234.

- [20] N. Zhang, Q. He, L. Zhang, J. Zhang, L. Huang, X. Yao, Homogeneous Fluorine Doping toward Highly Conductive and Stable Li₁₀Gep₂S₁₂ Solid Electrolyte for All-Solid-State Lithium Batteries, *Adv. Mater.* 36 (36) (2024) 2408903.
- [21] N. Zhang, X. Zhao, G. Liu, Z. Peng, J. Wu, M. Meng, X. Yao, Solid electrolyte membranes for all-solid-state rechargeable batteries, *eTransportation* (2024) 100319.
- [22] X. Zhao, L. Shen, N. Zhang, J. Yang, G. Liu, J. Wu, X. Yao, Stable binder boosting sulfide solid electrolyte thin membrane for all-solid-state lithium batteries, *Energy Mater. Adv.* 5 (2024) 0074.
- [23] K. Lee, S. Kim, J. Park, S.H. Park, A. Coskun, D.S. Jung, W. Cho, J.W. Choi, Selection of binder and solvent for solution-processed all-solid-state battery, *J. Electrochem. Soc.* 164 (9) (2017) A2075.
- [24] J. Ruhl, L.M. Riegger, M. Ghidui, W.G. Zeier, Impact of solvent treatment of the superionic argyrodite Li₆PS₅Cl on solid-state battery performance, *Advanced Energy and Sustainability, Research 2* (2) (2021) 2000077.
- [25] D.Y. Oh, Y.J. Nam, K.H. Park, S.H. Jung, S.J. Cho, Y.K. Kim, Y.G. Lee, S.Y. Lee, Y. S. Jung, Excellent compatibility of solvate ionic liquids with sulfide solid electrolytes: toward favorable ionic contacts in bulk-type all-solid-state lithium-ion batteries, *Adv. Energy Mater.* 5 (2015) 22.
- [26] J. Lee, T. Lee, K. Char, K.J. Kim, J.W. Choi, Issues and advances in scaling up sulfide-based all-solid-state batteries, *Acc. Chem. Res.* 54 (17) (2021) 3390–3402.
- [27] Y.J. Nam, D.Y. Oh, S.H. Jung, Y.S. Jung, Toward practical all-solid-state lithium-ion batteries with high energy density and safety: comparative study for electrodes fabricated by dry-and slurry-mixing processes, *J. Power Sources* 375 (2018) 93–101.
- [28] M. Yamamoto, Y. Terauchi, A. Sakuda, M. Takahashi, Binder-free sheet-type all-solid-state batteries with enhanced rate capabilities and high energy densities, *Sci. Rep.* 8 (1) (2018) 1212.
- [29] S. Kiritharan, S. Lucas, R. Degl'Innocenti, X. Hua, R. Dawson, H. Lin, Porosity characterisation of solid-state battery electrolyte with terahertz time-domain spectroscopy, *J. Power Sources* 595 (2024) 234050.
- [30] R. Amin, Y.-M. Chiang, Characterization of electronic and ionic transport in Li_{1-x}Ni_{0.33}Mn_{0.33}Co_{0.33}O₂ (NMC333) and Li_{1-x}Ni_{0.50}Mn_{0.20}Co_{0.30}O₂ (NMC523) as a function of Li content, *J. Electrochem. Soc.* 163 (8) (2016) A1512.
- [31] B. Marinho, M. Ghislandi, E. Tkalya, C.E. Koning, G. de With, Electrical conductivity of compacts of graphene, multi-wall carbon nanotubes, carbon black, and graphite powder, *Powder Technol.* 221 (2012) 351–358.
- [32] B.L. Trembacki, A.N. Mistry, D.R. Noble, M.E. Ferraro, P.P. Mukherjee, S. A. Roberts, Mesoscale analysis of conductive binder domain morphology in lithium-ion battery electrodes, *J. Electrochem. Soc.* 165 (13) (2018) E725–E736.
- [33] P.D. Desai, H. James, C.Y. Ho, Electrical resistivity of aluminum and manganese, *J. Phys. Chem. Ref. data* 13 (4) (1984) 1131–1172.
- [34] R.A. Matula, Electrical resistivity of copper, gold, palladium, and silver, *J. Phys. Chem. Ref. Data* 8 (4) (1979) 1147–1298.
- [35] J. Song, D.O. Shin, S. Byun, Y. Roh, C. Bak, J. Song, J. Choi, H. Lee, T.-S. Kwon, Y.-G. Lee, A new perspective on the advanced microblade cutting method for reliable adhesion measurement of composite electrodes, *J. Electrochem. Sci. Technol.* 13 (2) (2022) 227–236.
- [36] Y. SUKEUN, Crater-like architectural aluminum current collectors with superior electrochemical performance for Li-ion batteries, *J. Electroanal. 797* (2017) 2005–2017. Chemistry Elsevier Amsterdam, NL.
- [37] K. Kim, S. Byun, I. Cho, M.-H. Ryou, Y.M. Lee, Three-dimensional adhesion map based on surface and interfacial cutting analysis system for predicting adhesion properties of composite electrodes, *ACS Appl. Mater. Interfaces* 8 (36) (2016) 23688–23695.
- [38] B. Son, M.-H. Ryou, J. Choi, T. Lee, H.K. Yu, J.H. Kim, Y.M. Lee, Measurement and analysis of adhesion property of lithium-ion battery electrodes with SAICAS, *ACS Appl. Mater. Interfaces* 6 (1) (2014) 526–531.
- [39] G. Homann, P. Meister, L. Stolz, J.P. Brinkmann, J.R. Kulisch, T. Adermann, M. Winter, J. Kasnatscheew, High-voltage all-solid-state lithium battery with sulfide-based electrolyte: challenges for the construction of a bipolar multicell stack and how to overcome them, *ACS Appl. Energy Mater.* 3 (4) (2020) 3162–3168.
- [40] T. Graedel, J. Franey, G. Kammlott, The corrosion of copper by atmospheric sulphurous gases, *Corros. Sci.* 23 (11) (1983) 1141–1152.
- [41] M. Li, B. Wang, J. Ma, Z. Wang, Y. Liang, Z. Wang, L. Zhang, Y. Tang, Q. Huang, J. Huang, Understanding the stability of copper current collector with sulfide electrolyte in all-solid-state batteries, *Adv. Energy Mater.* 14 (7) (2024) 2303156.
- [42] D.S. Ko, J.H. Park, B.Y. Yu, D. Ahn, K. Kim, H.N. Han, W.S. Jeon, C. Jung, A. Manthiram, Degradation of high-nickel-layered oxide cathodes from surface to bulk: a comprehensive structural, chemical, and electrical analysis, *Adv. Energy Mater.* 10 (36) (2020) 2001035.
- [43] M. Lain, G. Apachitei, D.-E. Dogaru, W.D. Widanage, J. Marco, M. Copley, Measurement of anisotropic volumetric resistivity in lithium ion electrodes, *RSC Adv.* 13 (47) (2023) 33437–33445.
- [44] Y. Han, S.H. Jung, H. Kwak, S. Jun, H.H. Kwak, J.H. Lee, S.T. Hong, Y.S. Jung, Single- or poly-crystalline Ni-rich layered cathode, sulfide or halide solid electrolyte: which will be the winners for all-solid-state batteries? *Adv. Energy Mater.* 11 (21) (2021) 2100126.
- [45] H.H. Sun, H.-H. Ryu, U.-H. Kim, J.A. Weeks, A. Heller, Y.-K. Sun, C.B. Mullins, Beyond doping and coating: prospective strategies for stable high-capacity layered Ni-rich cathodes, *ACS Energy Lett.* 5 (4) (2020) 1136–1146.
- [46] D.Y. Oh, Y.E. Choi, Y.J. Nam, L. Han, J.-Y. Kim, H. Xin, F. Lin, S.M. Oh, Y.S. Jung, Solution-processable glass Li_{1-x}Li₄Sn₄ superionic conductors for all-solid-state Li-ion batteries, *Adv. Mater.* 8 (2015) (BNL-111621-2015-JA).
- [47] T. Asano, S. Yubuchi, A. Sakuda, A. Hayashi, M. Tatsumisago, Electronic and ionic conductivities of LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂-Li₃PS₄ positive composite electrodes for all-solid-state lithium batteries, *J. Electrochem. Soc.* 164 (14) (2017) A3960.
- [48] T.Y. Kwon, K.T. Kim, D.Y. Oh, Y.B. Song, S. Jun, Y.S. Jung, Three-dimensional networking binders prepared in situ during wet-slurry process for all-solid-state batteries operating under low external pressure, *Energy Storage Mater.* 49 (2022) 219–226.
- [49] E. Rossi, J.M. Wheeler, M. Sebastiani, High-speed nanoindentation mapping: a review of recent advances and applications, *Curr. Opin. Solid State Mater. Sci.* (2023) 101107.
- [50] W. Yan, C.L. Pun, Z. Wu, G.P. Simon, Some issues on nanoindentation method to measure the elastic modulus of particles in composites, *Compos. Part B: Eng.* 42 (8) (2011) 2093–2097.
- [51] J. Zhang, H. Zhong, C. Zheng, Y. Xia, C. Liang, H. Huang, Y. Gan, X. Tao, W. Zhang, All-solid-state batteries with slurry coated LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ composite cathode and Li₆PS₅Cl electrolyte: effect of binder content, *J. Power Sources* 391 (2018) 73–79.
- [52] F. Hippauf, B. Schumm, S. Doerfler, H. Althues, S. Fujiki, T. Shiratsuchi, T. Tsujimura, Y. Aihara, S. Kaskel, Overcoming binder limitations of sheet-type solid-state cathodes using a solvent-free dry-film approach, *Energy Storage Mater.* 21 (2019) 390–398.
- [53] N.C. Rosero-Navarro, T. Kinoshita, A. Miura, M. Higuchi, K. Tadanaga, Effect of the binder content on the electrochemical performance of composite cathode using Li₆PS₅Cl precursor solution in an all-solid-state lithium battery, *Ionics* 23 (2017) 1619–1624.
- [54] T. Liu, Q. Chu, C. Yan, S. Zhang, Z. Lin, J. Lu, Interweaving 3D network binder for high-area-capacity Si anode through combined hard and soft polymers, *Adv. Energy Mater.* 9 (3) (2019) 1802645.
- [55] G. Yoo, S. Kim, C. Chanthad, M. Cho, Y. Lee, Elastic rubber-containing multifunctional binder for advanced Li-S batteries, *Chem. Eng. J.* 405 (2021) 126628.
- [56] P. Zhu, D. Gastol, J. Marshall, R. Sommerville, V. Goodship, E. Kendrick, A review of current collectors for lithium-ion batteries, *J. Power Sources* 485 (2021) 229321.
- [57] B. Aktekin, L.M. Riegger, S.-K. Otto, T. Fuchs, A. Henss, J. Janek, SEI growth on Lithium metal anodes in solid-state batteries quantified with coulometric titration time analysis, *Nat. Commun.* 14 (1) (2023) 6946.
- [58] A. Pandey, S. Dalal, S. Dutta, A. Dixit, Structural characterization of polycrystalline thin films by X-ray diffraction techniques, *J. Mater. Sci.: Mater. Electron.* 32 (2021) 1341–1368.
- [59] Höltschi, L. Understanding Degradation Mechanisms of Sulfide-Based All-Solid-State Lithium-Ion Batteries at Multiple Length Scales. ETH Zurich, 2021.
- [60] D.H. Tan, E.A. Wu, H. Nguyen, Z. Chen, M.A. Marple, J.-M. Doox, X. Wang, H. Yang, A. Banerjee, Y.S. Meng, Elucidating reversible electrochemical redox of Li₆PS₅Cl solid electrolyte, *ACS Energy Lett.* 4 (10) (2019) 2418–2427.
- [61] Y.T. Chen, J. Jang, J.A.S. Oh, S.Y. Ham, H. Yang, D.J. Lee, M. Vicencio, J.B. Lee, D. H. Tan, M. Chouchane, Enabling Uniform and Accurate Control of Cycling Pressure for All-Solid-State Batteries, *Adv. Energy Mater.* (2024) 2304327.
- [62] J. Li, Y. Li, S. Zhang, T. Liu, D. Li, L. Ci, Long cycle life all-solid-state batteries enabled by solvent-free approach for sulfide solid electrolyte and cathode films, *Chem. Eng. J.* 455 (2023) 140605.
- [63] J. Li, Y. Li, S. Zhang, T. Liu, D. Li, L. Ci, In situ formed LiI interfacial layer for all-solid-state lithium batteries with Li₆PS₅Cl solid electrolyte membranes, *ACS Appl. Mater. Interfaces* 14 (50) (2022) 55727–55734.
- [64] S. Ito, S. Fujiki, T. Yamada, Y. Aihara, Y. Park, T.Y. Kim, S.-W. Baek, J.-M. Lee, S. Doo, N. Machida, A rocking chair type all-solid-state lithium ion battery adopting Li₂O–ZrO₂ coated LiNi_{0.8}Co_{0.1}Al_{0.1}O₂ and a sulfide based electrolyte, *J. Power Sources* 248 (2014) 943–950.
- [65] M. Grandjean, M. Perrey, X. Randrema, J. Laurier, P. Chenevier, C. Haon, S. Liatard, Low pressure cycling of solid state Li-ion pouch cells based on NMC–Sulfide–Nanosilicon chemistry, *J. Power Sources* 585 (2023) 233646.
- [66] S. Liu, L. Zhou, J. Han, K. Wen, S. Guan, C. Xue, Z. Zhang, B. Xu, Y. Lin, Y. Shen, Super long-cycling all-solid-state battery with thin Li₆PS₅Cl-based electrolyte, *Adv. Energy Mater.* 12 (25) (2022) 2200660.
- [67] K.T. Kim, T.Y. Kwon, Y.B. Song, S.-M. Kim, S.C. Byun, H.-S. Min, S.H. Kim, Y. S. Jung, Wet-slurry fabrication using PVDF-HFP binder with sulfide electrolytes via synergetic cosolvent approach for all-solid-state batteries, *Chem. Eng. J.* 450 (2022) 138047.



Hyeonseong Oh is a Ph.D. candidate under the co-supervision of Dr. Hun-Gi Jung at the Korea Institute of Science and Technology and Prof. Sang-Young Lee at the Yonsei University. His current research interests focus on the study of all-solid-state lithium batteries for the next generation secondary batteries



Sang-Young Lee, Professor of Chemical and Biomolecular Engineering at Yonsei University, Korea, received his BA in Chemical Engineering from Seoul National University (1991) and MS/PhD in Chemical Engineering from KAIST (1993, 1997). He served as a postdoctoral fellow at Max-Planck Institute and later led the development of ceramic-coated separators (SRS®) as a principal research scientist at Batteries R&D, LG Chem. He is a fellow of the Korean Academy of Science and Technology and a fellow of the National Academy of Engineering of Korea. His research focuses on liquid/solid electrolytes, high-mass-loading electrodes, cellulose-based batteries, and flexible/wearable power sources



Dr. Hun-Gi Jung is a principal research scientist in Korea Institute of Science and Technology. He received a Ph.D. degree in chemical engineering from Hanyang University, Republic of Korea in 2012. During his Ph.D., he worked on nanostructured anode materials for Li-ion batteries. His current research is mainly focused on electrode materials for Li-ion battery, Li-air battery, and all-solid-state battery as the next generation secondary batteries.